

Ratio Definitions

Reference guide to the 14 ratios calculated in *Financial_Health_Dashboard.xlsx*

Liquidity Ratios

Measure the company's ability to meet short-term obligations as they come due.

Current Ratio

$$\text{Total Current Assets} \div \text{Total Current Liabilities}$$

Shows how many dollars of current assets are available to cover each dollar of current liabilities. A ratio above 1.5–2.0x is generally considered healthy for a retail business.

Quick Ratio (Acid-Test)

$$(\text{Total Current Assets} - \text{Inventory}) \div \text{Total Current Liabilities}$$

A stricter liquidity test that excludes inventory, since inventory can be slow or difficult to convert to cash. A ratio near or above 1.0x is considered strong.

Cash Ratio

$$\text{Cash \& Equivalents} \div \text{Total Current Liabilities}$$

The most conservative liquidity measure, considering only cash and cash equivalents against current liabilities.

Profitability Ratios

Measure how effectively the company converts revenue and assets into profit.

Gross Margin

$$\text{Gross Profit} \div \text{Revenue}$$

The percentage of revenue remaining after subtracting the direct cost of goods sold, before operating expenses.

Operating Margin

$$\text{Operating Income (EBIT)} \div \text{Revenue}$$

The percentage of revenue remaining after both cost of goods sold and operating expenses, before interest and taxes.

Net Profit Margin

$$\text{Net Income} \div \text{Revenue}$$

The percentage of revenue that becomes bottom-line profit after all expenses, interest, and taxes.

Return on Assets (ROA)

$$\text{Net Income} \div \text{Total Assets}$$

Measures how efficiently the company uses its total asset base to generate profit.

Return on Equity (ROE)

$$\text{Net Income} \div \text{Total Shareholders' Equity}$$

Measures the return generated on shareholders' invested capital.

Leverage Ratios

Measure how much the company relies on debt financing and its ability to service that debt.

Debt-to-Equity

$$\text{Total Liabilities} \div \text{Total Shareholders' Equity}$$

Compares total debt and other obligations to the equity cushion available to absorb losses. A lower ratio indicates a more conservative capital structure.

Debt-to-Assets

$$\text{Total Liabilities} \div \text{Total Assets}$$

Shows what portion of total assets is financed by debt rather than equity.

Interest Coverage Ratio

$$\text{Operating Income (EBIT)} \div \text{Interest Expense}$$

Measures how many times over the company's operating income could cover its interest obligations. Higher is safer.

Efficiency Ratios

Measure how effectively the company uses its assets and working capital to generate revenue.

Asset Turnover

$$\text{Revenue} \div \text{Total Assets}$$

Measures how much revenue is generated per dollar of total assets — a broad measure of asset-use efficiency.

Inventory Turnover

$$\text{Cost of Goods Sold} \div \text{Inventory}$$

Shows how many times inventory is sold and replaced over the period. Higher turnover generally signals efficient inventory management.

Receivables Turnover

$$\text{Revenue} \div \text{Accounts Receivable}$$

Shows how many times receivables are collected over the period. Higher turnover signals faster collection of customer payments.

Note: Benchmark thresholds used to rate each ratio as Strong / Moderate / Weak are documented on the Benchmarks sheet of Financial_Health_Dashboard.xlsx, alongside the direction (higher-is-better vs. lower-is-better) applied to each ratio.